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Optimization of energy consumption for soybean production using Data Envelopment Analysis (DEA) approach

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ABSTRACT

In this study a non-parametric method of Data Envelopment Analysis (DEA) is used to estimate the energy efficiencies of soybean producers based on eight energy inputs including human labor, diesel fuel, machinery, fertilizers, chemicals, water for irrigation, electricity and seed energy and single output of grain yield. The study also helps to rank efficient and inefficient farmers and to identify optimal energy requirement and wasteful uses of energy. Data were collected using face-to-face surveys from 94 farms in Golestan province which is the most important center of soybean production in Iran. Based on the results, average yield and energy consumption for soybean production were 3233.15 kg ha^{-1} and 35372.23 MJ ha^{-1}, respectively. Also, the results of DEA application showed that, the technical, pure technical and scale efficiencies of farmers were 0.853, 0.919 and 0.926, respectively. Moreover, energy saving target ratio for soybean production was calculated as 20.12%, indicating that by following the recommendations resulted from this study, about 7116.84 MJ ha^{-1} of total input energy could be saved while holding the constant level of soybean yield. Also, electrical energy had the highest share (78.08%) from total saving energy, followed by fertilizers (10.46%) and diesel fuel (6.18%) energy inputs.

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1. Introduction

Soybean (*Glycine max* (*L.*) *Merrill*) is an annual legume that is primarily produced as a source of edible oil for human consumption. On the other hand, its residual meal is used as a supplemental protein source for feeding livestock; its crude protein content ranges from 41% to 50% (dry matter basis) [1,2]. Soybean is one of the most important oilseed crops cultivated in Iran. Iran produced about 209,000 tons of soybean grain from 115,000 hectares harvested land area in 2008 [3]. Golestan province is the main center of soybean production in Iran; about 75% from total soybean production in Iran is provided from this province [4].

Energy is one of the most important material bases for the economic growth and social development of a country or region. Scientific forecasts and analysis of energy consumption will be of great importance for the planning of energy strategies and policies. Nowadays, energy usage in agriculture has been intensified in response to continued growth of human population, tendency for an overall improved standard of living and limited supply of arable land; thus, the farmers use their inputs in excess and inefficiently, particularly when the inputs have low price or are available in plenty. The enhancement of energy efficiency not only helps in improving competitiveness through cost reduction, also results in minimized energy-related environmental pollution, thus positively contributing towards sustainable development [5,6].

There are several parametric and non-parametric approaches to measure the efficiency in agricultural productions; based on the literature, the indices of energy use efficiency (output energy to input energy ratio) and specific energy (i.e., input energy to yield ratio) have been used extensively to measure the energy efficiency of agricultural production systems [6–8]. Also, Singh et al. [9] investigated the technical efficiency for a sample of 382 randomly selected farms to establish the optimum level of energy input for wheat production in the different agro-climatic zones of Punjab. In this study a linear programming technique based on the concept of one-to-one functions was used.

Data Envelopment Analysis (DEA) is a non-parametric technique of frontier estimation. It determines the relative efficiency of a number of decision making units (DMUs) and target values for their improvement [10]. DEA allows the decision makers to simultaneously consider multiple inputs and outputs, when efficiency of each DMU is compared to that of an ideal operating unit rather than to the average performance [11,12]. Due to the high advantages of the data envelopment analysis, there is a large number of its application for evaluating the performance of DMUs in different issues [13–18]. Moreover in recent years, many authors have applied DEA in agricultural enterprises: Fraser and Cordina [19] used DEA to evaluate the technical efficiency of irrigated dairy farms in Australia. The results from this study proposed that DEA

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was a useful tool in helping to benchmark the dairy industry which is continually striving to improve the productive efficiency of farms. In another study, DEA was applied to investigate the efficiency of individual farmers and to identify the efficient units in citrus production in Spain [20]. Also, Barnes [21] identified the technical efficiency scores of Scottish dairy farms by applying the non-parametric method of DEA. Chauhan et al. [22] applied DEA approach to determine the efficiency scores of paddy producers with regard to six input parameters of human labor, diesel, seed, farmyard manure, fertilizers and machinery energy inputs and the paddy yield as output parameter. In another study by Nassiri and Singh [23], the DEA technique was subjected to the data of energy use for paddy production; also, the technical, pure technical and scale efficiency of farmers were estimated. Mousavi-Avval et al. [24] employed the DEA technique to benchmark and rank the productivity performance of apple producers based on the amount of various energy inputs and output of apple yield values.

Based on the literature, there was no study on optimization of energy inputs for soybean production in Iran. So, the objectives of this research were to specify energy use pattern for soybean production, analyze the efficiencies of farmers, rank efficient and inefficient ones and to identify target energy requirement and wasteful uses of energy from different inputs for soybean production in Golestan province of Iran.

2. Materials and methods

2.1. Selection of case study region and data collection

This study was conducted in Golestan province of Iran. This province is located within 36° 30′ and 38° 08′ north latitude and 53° 57′ and 56° 22′ east longitude. The selection of this province as the case study region allows more validity to the assumptions of DEA that all farmers operate in a relatively homogenous region, and hence, technical efficiency of farmers will not be adversely affected through climatic and bio-physical constraints.

Data were collected through personal interview method in a specially designed schedule for this study. The collected data belonged to the 2009/10 production year. Before collecting data, a pre-test survey was conducted by a group of randomly selected farmers. The required sample size was determined using simple random sampling method. The equation is as below [5]:

$$n = \frac{N \times s^2 \times t^2}{(N-1)d^2 + (s^2 \times t^2)} \tag{1}$$

where n is the required sample size, N is the number of soybean producers in target population, s is the standard deviation in the pre-tested data, t is the t value at 95% confidence limit (1.96), and d is the acceptable error. The permissible error in the sample size was defined to be 5% for 95% confidence. Thus the sample size was found to be 94. Consequently, based on the number of soybean producers in each village the 94 farmers from the population were randomly selected. Totally, the soybean producers in twelve villages in the study region were surveyed.

2.2. Energy equivalents of inputs and output

The data included the quantity of various energy inputs used per hectare of soybean production including: human labor, machinery, diesel fuel, chemicals, fertilizer, water for irrigation, seed and electricity, and the production yield as output. In order to analyze the performance of farmers from an energy use efficiency point of view, all of inputs and output were then converted into energy equivalents by multiplying the quantity of input use with their corresponding energy equivalent coefficients. The energy coefficients of inputs are the energy used from primary production to the end user. The energy equivalents of some inputs may vary in different countries. There was no study to calculate the energy equivalents of inputs in Iranian condition; therefore in this study the coefficients were obtained from the previous studies conducted in similar conditions as presented in Table 1.

The energy equivalent of human labor is the muscle power used in field operations of crop production. Chemicals and chemical fertilizers energy equivalents means the energy consumption for producing, packing and distributing the materials and they are given on an active ingredient basis. Farmyard manure is regarded as a source of nutrients, so the energy equivalent of farmyard manure equates with that of mineral fertilizer equivalents corresponding to the fertilization effect of the applied manure. Also, the energy sequestered in diesel fuel and electricity means their heating value (enthalpy) and the energy needed to make their energy available directly to the farmers. Moreover, the seed energy is the energy used in the production of a crop and the grain energy is the gross energy content determined from laboratory bomb calorimeter tests [27,30].

The energy equivalent of water for irrigation input means indirect energy of irrigation consist of the energy consumed for manufacturing the materials for the dams, canals, pipes, pumps, and equipment as well as the energy for constructing the works and building the on-farm irrigation systems [31].

For calculating the embodied energy in agricultural machinery it was assumed that the energy consumed for the production of the tractors and agricultural machinery be depreciated during their economical life time [29]; therefore, the machinery energy input was calculated using the following Eq. [32]:

$$ME = \frac{G \times M_P \times t}{T} \tag{2}$$

where ME is the machinery energy per unit area (MJ ha^{-1}); G is the machine mass (kg); M_P the production energy of machine (MJ kg^{-1}); t is the time that machine used per unit area (h ha^{-1}) and T is the economic life time of machine (h).

2.3. Data envelopment analysis technique

The DEA is an analysis method to measure the relative efficiency of a homogeneous number of organizations that essentially perform the same tasks [33]. In this case, they are soybean production farms. So, the values of energy consumed from different energy inputs (MJ ha^{-1}), as mentioned above, were defined as input parameters, and the yield of soybean grain production (kg ha^{-1}) was defined as output parameter; also, each farmer was called a DMU. The data analysis was carried out with help of the Frontier Analyst software.

In DEA, an inefficient DMU can be made efficient either by reducing the input levels while holding the outputs constant (input oriented); or symmetrically, by increasing the output levels while holding the inputs constant (output oriented) [13]. The choice between input and output orientation depends on the unique characteristics of the set of DMUs under study. In this study the input oriented approach was deemed to be more appropriate because there is only one output while multiple inputs are used; also as a recommendation, input conservation for given outputs seems to be a more reasonable logic [18]; so the soybean production yield is hold fixed and the quantity of source wise energy inputs was reduced.

DEA has two models including CCR and BCC models. The CCR DEA model developed by Charness et al. [34] and assumes constant returns to scale. It measures the technical efficiency by which the DMUs are evaluated for their performance relative to other DMUs in a sample [35]. On the other hand the BCC DEA model develoed

Table 1
Energy equivalent coefficients of inputs and output in soybean production in Golestan, Iran.

Item	Unit	Energy equivalent (MJ unit^{-1})	References
A. Inputs			
1. Human labor	h	1.96	[25]
2. Machinery	kg		
a. Tractor		93.61	[26]
b. Self propelled combine		87.63	[26]
c. Other machinery		62.7	[26]
3. Diesel fuel	L	47.8	[27]
4. Chemicals	kg		
a. Herbicides		238	[25]
b. Insecticides		101.2	[25]
5. Fertilizer	kg		
a. Nitrogen		66.14	[28]
b. Phosphate (P_2O_5)		12.44	[28]
c. Potassium (K_2O)		11.15	[28]
d. Sulfur (S)		1.12	[25]
e. Farmyard manure		0.3	[25]
6. Water for irrigation	m^3	1.02	[25]
7. Seed	kg	3.6	[29]
8. Electricity	kWh	11.93	[28]
B. Output			
1. Soybean	kg	25	[29]

by Banker et al. [36] and assumes variable returns to scale conditions. It decomposes the technical efficiency into pure technical efficiency for management factors and scale efficiency for scale factors. Thus, pure technical efficiency is the technical efficiency that has the effect of scale efficiency removed [24]. Scale efficiency gives quantitative information of scale characteristics; it is the potential productivity gain from achieving optimal size of a DMU. Scale efficiency can be calculated by the relation between technical and pure technical efficiencies as below [23]:

$$\text{Scale efficiency} = \frac{\text{Technical efficiency}}{\text{Pure technical efficiency}} \tag{3}$$

The concept of three types of efficiencies is graphically illustrated in Fig. 1, in which, the straight line of MN that passes through the origin and the extreme data point presents the envelope of the data set with constant returns to scale; while, the envelope of the data set with variable returns to scale is depicted by the piecewise line joining P_1, P_2, P_3 and P_4. The DMU situated on the line MN is considered as efficient and has a technical efficiency score of 1. Also, all the DMUs situated on the piecewise line (P_1, P_2, P_3 and P_4) have a pure technical efficiency score of 1. Finally, the scale efficiency for DMUs which has both technical and pure technical efficiency score equal to 1, is unity, such as P_2; whereas for the other DMUs, it is less than one.

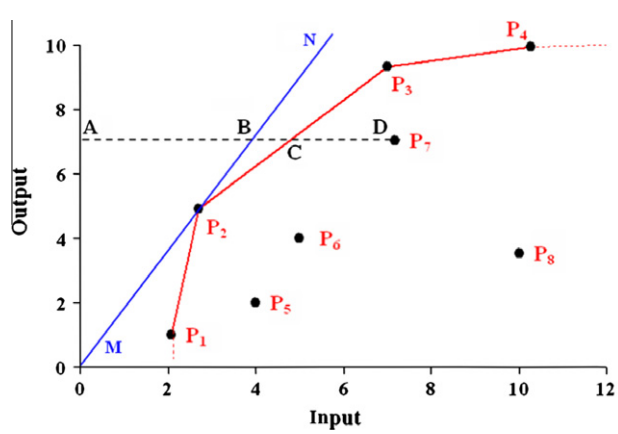

Fig. 1. Geometric interpretation of different efficiencies [22].

Considering the DMU 7 (P_7) in Fig. 1, its input and output are given by AD and MA, respectively. B and C are the points of intersection of the line AD with the line MN and the piecewise line of the envelope of the data set. One can interpret that, AB is the ideal input required to produce the output B on MN, if constant return to scale is to prevail. However, considering variable return to scale, one can relax the input requirement to be equal to AC to be able to produce the output B on MN. One can now define the various efficiencies as follow [22]:

$$\text{Pure Technical Efficiency} = AC/AD \tag{4}$$

$$\text{Technical Efficiency} = AB/AD \tag{5}$$

$$\text{Scale Efficiency} = AB/AC \tag{6}$$

The results of standard DEA models separate the DMUs into two sets, those that are efficient and locate on the frontier line and those that are inefficient; so many units are calculated as efficient and can not be ranked. In order to rank all the DMUs, discrimination among the efficient farmers is necessary, because in DEA it is possible that some of the efficient units have better performance than the other efficient ones. This is an additional reason for the growing interest in complete ranking techniques; to overcome this problem, a number of methods such as cross efficiency, super efficiency, benchmarking method, etc. have been suggested in the literature [37]. In this study, the cross efficiency ranking method, developed by Sexton et al. [38], was used. This method is a commonly used approach for ranking the efficient DMUs and has previously been used for ranking the efficient farmers in some agricultural enterprises such as apple [24] and paddy [22] productions. In this method, the results of all the DEA efficiency scores can be aggregated in a matrix, called cross efficiency matrix, in which, E_{ij}, the element in the ith row and jth column, represents the efficiency score for the jth farmer calculated using the optimal weights of the ith farmer which is computed by the CCR model. In general, the efficient farmers can be ranked according to their average cross efficiency score which can be achieved by averaging each column of cross efficiency matrix and it is a matter of judgment for analysis to select the highly ranked farmers as truly efficient ones; so, a farmer with a high average cross efficiency score is a good performer [11,12,22].

In the analysis of efficient and inefficient DMUs, the energy saving target ratio (ESTR) index was used as follows [39]:

$$ESTR_j(\%) = \frac{(\text{Energy Saving Target})_j}{(\text{Actual Energy Input})_j} \times 100 \tag{7}$$

where energy saving target is the total reducing amount of input that could be saved without decreasing output level and j represents the jth DMU. ESTR represents the inefficiency level for each DMU with respect to energy use. The minimal value of energy saving target is zero, so the percentage of ESTR will be between zero and 100. A zero ESTR percentage shows the DMU on the frontier, such as efficient ones; on the other hand for inefficient DMUs, the ESTR percentage is larger than zero and means that energy could be saved. A higher ESTR percentage implies higher energy inefficiency and a higher energy saving amount [39].

3. Results and discussion

3.1. Analysis of energy input and output in soybean production

Table 2 presents the characteristics and used time of different agricultural machinery, employed in soybean production operations. In the last column, the total energy equivalent from each machine is presented. As it is seen, the total machinery energy input for soybean production was calculated as 962.89 MJ ha^{-1}.

The quantity of other physical inputs used in the production of soybean, output rate and their energy equivalents are presented in Table 3. The results revealed that, human labor used in soybean production was as 194.55 h ha^{-1}; it was mainly used for irrigation (27.87%), weeding (25.40%) and harvesting (15.50%) operations. Also, 5.33 kg chemicals, 157.88 kg chemical fertilizer and 5347.87 kg farmyard manure per hectare of soybean production were used in surveyed region. The quantities of other inputs consumed per unit area of soybean production were: 103.61 L diesel fuel, 3302.96 m^3 water for irrigation, 68.80 kg seed and 1335.94 kWh electricity. The average soybean output was found to be 3233.15 kg ha^{-1}.

The energy equivalent of inputs and output in soybean production are also presented in Table 3. It is evident that, the total energy input for soybean production was about 35,372 MJ ha^{-1}, from which, the highest share was consumed by electrical energy, followed by fertilizers and diesel fuel inputs. On the other hand, the output energy was equal to 80828.75 MJ ha^{-1} in the farms that were studied. In the last column of Table 3 the standard deviation values for the inputs and output used in DEA model are presented. Sheikh Davoodi and Houshyar [40] investigated the energy consumption for canola and sunflower production in Iran. They reported that, total energy requirements for canola and sunflower were about 30,889 and 22,945 MJ ha^{-1}, respectively. Also in both the crops, fertilizer, electricity and diesel fuel were the three main energy consuming inputs.

3.2. Efficiency estimation of farmers

The results of BCC and CCR DEA models are presented in Fig. 2. The results revealed that, from the total of 94 farmers considered for the analysis, 40 farmers (42.55%) had the pure technical efficiency score of 1. Moreover, from the pure technically efficient farmers 26 farmers (27.66%) had the technical efficiency score of 1; indicating that they were globally efficient and were operating at the most productive scale size of production; however, the remainder of 14 pure technically efficient farmers were only locally efficient ones; it was due to their disadvantageous conditions of scale size. On the other hand, from inefficient farmers, 19 farmers had their pure technical efficiency in the 0.9–0.99 range; while 21 farmers were in the efficiency range of 0.8–0.89.

The summarize statistics for the three estimated measures of efficiency are given in Table 4. The results revealed that on average, the technical, pure technical and scale efficiency scores were 0.853, 0.919 and 0.926, respectively. Moreover, the technical efficiency varied from 0.583 to 1, with the standard deviation of 0.132 which had the highest variation between other efficiency scores. The wide variation in the technical efficiency of farmers indicated that all the farmers were not fully aware of the right production techniques or did not applied them at the proper time, in the optimal quantity.

Nassiri and Singh [23] applied the non-parametric method of DEA to determine the technical, pure technical and scale efficiencies of farmers in paddy production in Punjab; they reported that in zone 2, technical, pure technical and scale efficiency scores were as 0.88, 0.91 and 0.96, respectively. In another study, the technical and pure technical efficiencies of farmers were reported about 0.79 and 0.90, respectively [24].

3.3. Ranking the efficient farmers

In this study for ranking the farmers operating on the efficiency frontier, the cross efficiency scores based on the results of CCR model in each cell of the cross efficiency matrix were calculate. The average and standard deviation of cross efficiency scores for 10 superior efficient farmers are presented in Table 5. The results revealed that, farmers with Nos. 2, 45, 83 and 28 with the average cross efficiency scores of 0.661, 0.654, 0.653 and 0.651 had the highest average cross efficiency scores, respectively; so, these farms can be used as terms of benchmarking and establishing the best practice management.

3.4. Comparing input use pattern of efficient and inefficient farmers

The quantity of energy inputs from different sources and output for 10 superior efficient farmers and inefficient ones (based on the CCR model) are given in Table 6. The results revealed that efficient farmers used all of physical inputs less than inefficient farmers. The use of electricity, farmyard manure and water for irrigation by efficient farmers were less than inefficient ones by 61.48%, 35.27% and 28.59%, respectively. Also, the production yield for efficient farmers was found to be 12.14% higher than that of inefficient ones.

3.5. Energy saving from different energy inputs

Table 7 shows the optimum energy requirement and saving energy for soybean production based on the results of BCC model. Also, the percentage of ESTR are illustrated in the last column. As indicated, optimum energy requirement for electricity

Table 2
Energy equivalents of machinery used in soybean production in Golestan, Iran.

Machinery	Weight (kg)	Life time (h)	Used time per unit area (h ha^{-1})	Total energy equivalent (MJ ha^{-1})
1. Tractor	3000	10,000	12.89	361.97
2. Self propelled combine	6000	2000	1.48	389.08
3. Other machinery				211.84
a. Cultivator	500	2500	1.04	13.04
b. Disk harrow	575	2500	2.01	28.99
c. Bund former	200	1200	0.54	5.64
d. Planter	600	1200	1.25	39.19
e. Fertilizer spreader	250	1200	0.23	3
f. Sprayer	400	1200	4.04	86.44
g. Trailer	750	5000	3.78	35.54
Total				962.89

Table 3
Amounts of inputs, output and their energy equivalents for soybean production.

Inputs (Unit)	Quantity (Unit ha^{-1})	Total energy equivalent (MJ ha^{-1})	SD[a]
A. Inputs			
1. Human labor (h)	194.55	381.32	101.97
2. Machinery (h)		962.89	246.95
3. Diesel fuel (L)	103.61	4952.56	1365.03
4. Chemicals (kg)	5.33	901.92	325.64
a. Herbicides	2.65	630.70	
b. Insecticides	2.68	271.22	
5. Fertilizer (kg)	5505.75	8619.08	3018.48
a. Nitrogen	94.97	6281.32	
b. Phosphate (P_2O_5)	50.43	627.35	
c. Potassium (K_2O)	9.18	102.36	
d. Sulfur (S)	3.30	3.70	
e. Farmyard manure	5347.87	1604.36	
6. Water for irrigation (m^3)	3302.96	3369.02	1206.19
7. Seed (kg)	68.80	247.68	37.04
8. Electricity (kWh)	1335.94	15937.76	9712.28
Total energy input		35372.23	
B. Output			
1. Soybean (kg)	3233.15	80828.75	492.28
Total energy output		80828.75	

[a] Indicates standard deviation for energy inputs (MJ ha^{-1}) and yield (kg ha^{-1}).

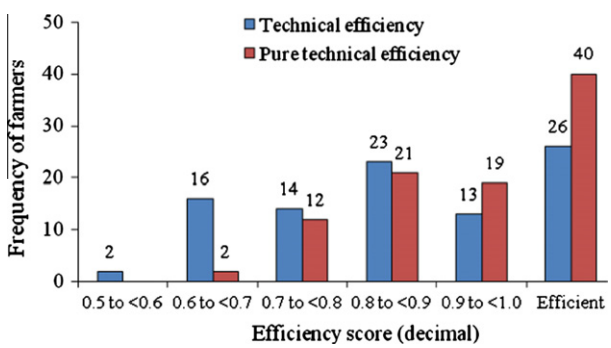

Fig. 2. Efficiency score distribution of soybean producers in Iran.

Table 4
Average efficiencies of farmers in soybean production.

Particular	Average	SD	Min	Max
Technical efficiency	0.853	0.132	0.583	1
Pure technical efficiency	0.919	0.095	0.638	1
Scale efficiency	0.926	0.088	0.681	1

Table 5
Average cross efficiency score for 10 superior efficient farmers.

Farmer no.	Cross efficiency	SD	Farmer No.	Cross efficiency	SD
2	0.661	0.253	75	0.635	0.183
45	0.654	0.242	81	0.628	0.254
83	0.654	0.225	7	0.626	0.252
28	0.651	0.255	86	0.612	0.206
1	0.644	0.265	32	0.610	0.246

(10381.15 MJ ha^{-1}) was found to be the highest. Also, the optimum values of fertilizer, diesel fuel, water for irrigation, machinery, chemicals, human labor and seed energy inputs were calculated as 7874.68, 4512.76, 3076.58, 929.73, 878.18, 358.39 and 243.92 MJ ha^{-1}, respectively. Moreover, the results of ESTR calculation showed that, 34.86% from electricity, 8.88% from diesel fuel, 8.68% from water for irrigation, 8.64% from fertilizer, 6.01% from human labor, 3.44% from machinery, 2.63% from chemicals and 1.52% from seed energy consumption could be saved. The percentage of total saving energy in optimum requirement over total actual use of energy was calculated as 20.12%, indicating that by following the recommendations resulted from this study, on average, about 7116.84 MJ ha^{-1} of total input energy could be saved while holding the constant output level of soybean yield.

In the region, electricity was mainly used in order to pump water for irrigation by electric motor. The high contribution of saving electrical energy resulted from the low efficiency of energy conversion by electric motor. Also, some of farmers have used flood irrigation and lack of good leveling of farms leaded to waste a lot of water and energy in the form of electricity. The high contribution of fertilizer energy inputs showed that, all of farmers were not fully aware of proper time and quantity of fertilizers usage. The high percent saving in diesel fuel shows the mismanagement in machinery employment in field operations. Also, it was due to the low prices and free availability to these inputs in surveyed region.

Singh et al. [9] in the study on optimization of energy use for wheat crop in zone 1 of Punjab reported that the electricity, fertilizers and diesel fuel energy inputs could be saved by 35.5%, 24% and 22.8%, respectively; also, the existing level of productivity could be achieved by 22.3%, 20.8%, 9.8%, 7.1% and 15.9% reducing the energy input over the actual energy input, in zones 1, 2, 3, 4 and 5, respectively. In another study, Chauhan et al. [22] reported that on an average, about 11.6% of the total input energy for paddy production in India could be saved.

Fig. 3 shows the share of the various energy inputs in the total input saving energy. It is evident that, the highest contribution to the total saving energy is 78.08% from electricity followed by fertilizer (10.46%), diesel fuel (6.18%) and water for irrigation (4.11%) energy inputs. Moreover, the contributions of machinery, chemicals, human labor, and seed energy inputs were relatively low. These inputs contributed to the total saving energy at less than 1%, showing that, they have been used in the right proportions by almost all the DMUs. The results indicate that there is a greater scope to increase the energy use efficiency by accurate use of electricity, fertilizers and diesel fuel energy inputs.

Table 6
Amounts of physical inputs and output for 10 superior efficient farmers and inefficient farmers.

Inputs (unit)	10 Superior efficient farmers (Unit ha^{-1}) (A)	Inefficient farmers (Unit ha^{-1}) (B)	Difference (%) $(B-A)*100/B$
A. Inputs			
1. Human labor (h)	173.37	199.4	13.05
2. Machinery (h)	21.79	28.86	24.5
3. Diesel fuel (L)	84.93	108.25	21.54
4. Chemicals (kg)			
a. Herbicides	2.35	2.85	17.54
b. Insecticides	1.55	2.76	43.84
5. Fertilizer (kg)			
a. Nitrogen	79.2	99.93	20.74
b. Phosphate (P_2O_5)	43.7	50.1	12.77
c. Potassium (K_2O)	7.5	7.54	0.53
d. Sulfur (S)	2.7	2.71	0.37
e. Farmyard manure	3931.25	6073.7	35.27
6. Water for irrigation (m^3)	2499.5	3500.34	28.59
7. Seed (kg)	62.50	70.22	10.99
8. Electricity (kWh)	572.51	1486.23	61.48
B. Output			
C. Soybean (kg)	3530	3147.79	-12.14

Table 7
Energy requirement in optimal condition and saving energy for soybean production.

Input	Optimal energy requirement (MJ ha^{-1})	Saving energy (MJ ha^{-1})	ESTR (%)
Human labor	358.39	22.93	6.01
Machinery	929.73	33.16	3.44
Diesel fuel	4512.76	439.8	8.88
Chemicals	878.18	23.74	2.63
Fertilizer	7874.68	744.4	8.64
Water for irrigation	3076.58	292.44	8.68
Seed	243.92	3.76	1.52
Electricity	10381.15	5556.61	34.86
Total	28255.39	7116.84	20.12

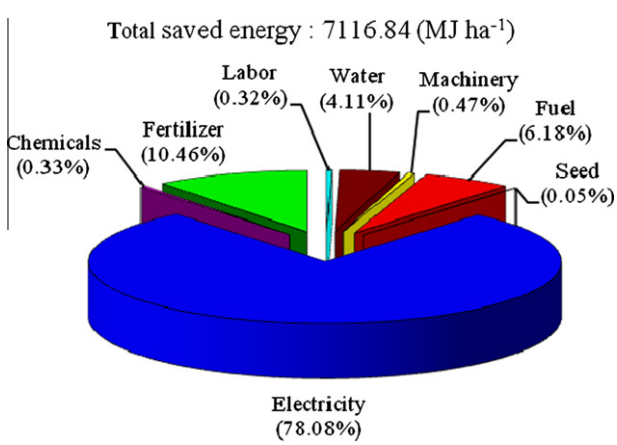

Fig. 3. Distribution of saving energy from different sources for soybean production in Iran.

Chauhan et al. [22] reported that the contribution of fertilizer and diesel energy inputs from total saving energy in paddy production were 33% and 24%, respectively. Mousavi-Avval et al. [24] reported that in optimization of energy use for apple production, the contribution of electrical energy from total saving energy was the highest. Also, fertilizer and diesel fuel had the relatively high contributions.

3.6. Setting realistic input levels for inefficient farmers

In Table 8 the pure technical efficiency, actual energy use and optimum energy requirement from different energy inputs for 54 individual inefficient farmers are presented. Using this information, it is possible to advise a producer regarding the better operating practices by following his/her target energy requirement from different inputs to reduce the input energy levels to the target values while achieving the output level presently achieved by him. It gives the average energy usage in current and optimal condition (MJ ha^{-1}) and percent contribution of total saving energy over actual use. Therefore, dissemination of these results will help to improve efficiency of farmers for soybean production in surveyed region. As it is evident in the last column of Table 8, for inefficient farmers, ESTR percentages is changes from 2.85% (farmer No. 69) to 61.55% (farmer no. 93), with the average of 28.42%. Indicating that between inefficient farmers with numbers 69 and 93 are the best and worst inefficient farmers, respectively.

To sum it up, soybean is an oilseed crop with high requirements on energy inputs especially electricity, fertilizer and diesel fuel energies. The results revealed that, there was a great potential for improving energy efficiency of soybean production in the region. Some strategies such as providing better extension services and farmer training programs and use of advanced technologies should be developed in order to increase the technical efficiency of soybean production in the region. The farmers should be trained with regard to the optimal use of inputs such as fertilizer, chemicals and irrigation water as well as production technologies. The

Table 8
The actual energy use and optimum energy requirements for individual inefficient soybean producers based on the results of BCC model.

DMU No.	PTE*	Actual energy use (MJ ha^{-1})								Optimal energy requirement (MJ ha^{-1})								ESTR (%)
		Water for irrigation	Human labor	Machinery	Diesel fuel	Electricity	Fertilizer	Chemicals	Seed	Water for irrigation	Human labor	Machinery	Diesel fuel	Electricity	Fertilizer	Chemicals	Seed	
3	0.849	3427	387	798	4229	23,303	7958	577	252	3154	369	798	4229	14,337	7231	577	252	24.39
4	0.837	2776	497	1082	5809	15,336	10,098	916	252	2776	388	1082	5610	10,137	10,098	916	252	14.98
5	0.943	2513	270	1022	5311	17,089	6112	729	360	2513	270	898	4606	12,398	6112	729	241	16.88
6	0.945	2468	298	947	4671	0	8448	916	216	2468	289	906	3585	0	6961	916	216	14.61
8	0.867	4935	418	986	5234	16,778	10,867	440	360	3968	418	986	5110	16,778	10,867	440	287	2.91
10	0.957	2879	269	976	4600	0	5566	1068	252	2487	269	858	3420	0	5566	811	216	12.71
12	0.939	2468	362	1013	6001	16,778	7018	1169	216	2468	295	1013	4253	4482	7018	935	216	40.96
15	0.772	4113	392	1320	7190	20,972	11,085	866	288	3557	392	1320	7094	16,686	11,085	866	288	10.68
16	0.757	3599	567	1075	5488	25,997	11,221	866	270	3599	466	1075	5059	14,301	10,253	866	270	26.88
17	0.922	3085	553	1161	6202	20,972	9439	967	216	3085	427	982	4513	7905	9439	967	216	35.36
18	0.834	2673	410	763	3947	18,176	7885	815	288	2673	369	760	3947	3607	7044	815	268	44.27
19	0.701	7403	525	1181	5711	34,604	9835	1524	288	5354	494	1181	5140	19,823	9835	1365	288	28.8
22	0.903	4627	350	974	5190	13,763	5764	1241	288	3677	350	974	5050	13,763	5764	1110	271	3.84
23	0.796	2879	392	1264	6241	9787	9304	1068	252	2879	328	1201	4837	3288	8525	1042	252	28.33
24	0.918	2056	481	949	5076	13,982	7451	866	288	2056	291	949	4937	10,651	7451	866	241	11.9
26	0.786	5552	515	1576	8358	25,953	9835	1068	252	3985	417	1303	5450	8501	8866	1061	252	43.82
27	0.786	5552	559	1793	9725	25,953	13,585	967	252	3658	364	1339	5396	5668	8649	967	252	54.97
33	0.823	3599	312	1215	6125	24,468	10,098	967	252	2947	312	1159	5255	7826	8957	967	252	41.16
37	0.868	4113	344	1050	5142	13,982	5430	916	252	3451	344	1050	4581	8221	5430	916	252	22.36
41	0.638	6169	568	1193	5585	41,945	14,174	1321	324	5344	568	1193	5585	22,800	13,727	1321	324	28.64
43	0.917	4113	687	862	5720	18,351	12,539	1458	216	3443	530	844	4606	11,275	12,266	1136	216	21.91
44	0.880	3085	260	1031	4778	13,763	7208	1372	270	2462	260	1031	4778	8896	7208	1059	246	18.34
46	0.804	2879	308	1109	5163	9787	8973	1018	252	2879	308	1109	4414	4419	7724	1018	252	24.98
47	0.865	2879	468	1057	5719	19,574	15,390	1078	252	2879	365	1057	5697	12,181	10,013	1078	252	27.78
48	0.959	3427	334	850	4251	10,195	9339	866	216	3249	334	850	4138	8249	8910	866	216	9.04
49	0.720	3173	431	1475	6978	18,875	13,075	1119	288	3173	376	1398	6324	10,867	10,274	1119	288	25.53
50	0.982	4935	542	1044	5615	33,556	7848	753	216	2956	374	1044	5507	12,175	7848	753	216	43.36
51	0.907	3011	365	1402	7897	21,369	12,514	661	234	3001	365	1299	6704	12,772	8681	661	234	28.95
53	0.982	4935	204	1051	5947	25,167	9349	1018	216	4338	204	748	3313	638	9026	1018	216	59.28
54	0.962	2570	333	763	3566	17,477	6085	916	216	2570	308	763	3424	2844	6085	842	216	46.59
56	0.984	2154	283	855	5050	12,816	8298	867	216	2154	283	855	4528	11,433	7921	867	216	7.47
59	0.895	3701	287	1114	6414	22,021	10,826	967	252	2704	287	1014	5442	13,376	9732	967	252	25.91
60	0.820	4750	317	1293	7018	28,260	10,826	967	252	3219	317	1180	5398	8223	9073	967	252	46.67
61	0.794	3599	480	1063	5684	16,822	10,240	1018	252	3514	447	1063	4872	9706	10,240	1018	252	20.55
63	0.867	3016	367	700	3771	23,070	10,492	729	252	2811	352	700	3746	8693	6823	729	252	43.14
64	0.927	2938	319	759	4225	17,477	8934	1018	252	2938	319	759	3890	12,503	8934	1018	249	14.79
66	0.830	3393	384	832	4915	25,953	10,492	967	252	3034	384	832	4220	7235	8162	967	252	46.84
67	0.983	2193	348	868	4759	0	8298	866	216	2193	312	779	3739	0	8289	866	216	6.58
68	0.819	2742	419	1046	6309	16,312	10,098	1137	252	2742	372	1046	5219	9302	9794	1137	252	22.05
69	0.914	2570	332	844	4375	15,292	8913	884	252	2570	332	844	4348	14,733	8544	884	252	2.85
70	0.841	2513	408	855	4990	17,089	15,852	1018	252	2513	335	855	4249	4047	8509	988	252	49.39
71	0.823	4113	324	1182	5938	20,972	9566	916	252	3106	324	1157	5283	6616	9189	916	252	37.96
72	0.799	3599	512	1087	5937	25,997	11,221	866	252	3416	407	1087	4784	11,460	9264	866	252	36.25
73	0.833	3085	554	1250	6578	20,972	7219	866	252	3085	425	1250	5900	10,635	7219	866	252	27.33
74	0.799	3342	328	1156	6517	22,720	8335	815	252	3223	328	1143	4782	8224	7831	815	252	38.8
76	0.897	3856	304	1211	6161	12,289	7564	967	252	2658	304	1120	5892	12,289	7564	967	246	4.8
77	0.841	2570	302	950	4596	12,015	10,098	916	252	2570	302	950	4579	7783	7806	916	252	20.64
78	0.749	6580	345	1236	5689	27,963	6663	967	288	4747	345	1222	4938	11,899	6663	931	288	37.6
79	0.829	3701	382	741	4235	22,021	11,729	1321	288	3701	382	741	3830	5627	9688	1149	288	42.81
82	0.663	3427	546	954	5531	10,923	16,513	1407	324	3427	508	954	5145	5261	12,277	1407	324	26.05
85	0.991	2313	277	1096	5036	0	12,653	628	216	2313	277	894	3892	0	7903	628	209	27.47
90	0.899	4113	335	737	3298	27,963	6085	815	252	2868	335	658	3298	5646	6085	815	249	54.23
91	0.781	6580	377	1040	4175	3915	10,405	815	270	3945	377	1040	4175	3915	7992	815	270	18.3
93	0.983	3701	389	794	3420	25,167	4805	916	216	2599	317	720	3420	2261	4805	816	216	61.55

* Pure technical efficiency (PTE).

local agricultural institutes in the region have an important role to inform the farmers with respect to more efficient use of energy resources and providing more sustainable agricultural production systems in the region.

4. Conclusions

In this research, the energy requirements of inputs and output for soybean production were examined in Golestan province, Iran. The results revealed that soybean production depends mainly on electricity, fertilizers and diesel fuel energy inputs. Consequently, the input oriented BCC and CCR DEA models were used to estimate the energy efficiencies of farmers. Based on the results of this study, electrical energy had the highest potential for improvement, followed by fertilizer and diesel fuel energy inputs. Also, about 20% potential of reduction in energy input utilization could be achieved if all soybean farms are operated efficiently. The reduction in wasteful uses of energy may even enhance the viability of soybean farms, giving farmers a more control over energy consumption. Improving efficiency of electric motors, providing educational opportunities to inefficient farmers for changing their wrong behaviors and also employing the controlled input use by performance monitoring and assessment studies can be suggested to improve energy efficiency. Finally, employing non-parametric method of DEA is confirmed as a very useful tool for benchmarking and improving the energy efficiency in agricultural production. The use of this methodology provides an important knowledge about the wasteful uses of energy by inefficient units.

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